MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY

MALAWI INSTITUTE OF TECHNOLOGY

MATH 2101: DISCRETE MATHEMATICS

ASSIGNMENT 1

STUDENTS NAMES - GROUP 13

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Question 1

(a) State the Pigeon Hole Principle. (**2 marks**)

Solution

The pigeon Hole Principle states that if p items are put into q containers, with $p > q$, then at least one container must contain more than one item.

(b) Out of 199 students who registered for Discrete Mathematics module at Malawi University of Science and Technology, how many students were born in the same month? (**3 marks**)

Solution

The number of Students born in the same month

$$= \left(\frac{P}{q}\right) \text{ where } P \text{ is pigeons (199 students) and } q$$
$$\text{is 12 months in a year}$$
$$= \frac{199}{12}$$
$$= 16.58$$

At least 17 Students were born in the same month.

(c) Find the minimum number of students in a class to be sure that at least two of them have last names that begin with the same letter. (**2 marks**)

Solution

$$\begin{aligned} \text{Holes } (q) &\text{ 26 letters in the alphabet} \\ &= q + 1 \text{ students} \\ &= 26 + 1 \\ &= \underline{\underline{27 \text{ students}}} \end{aligned}$$

(d) What is the minimum number of students, each of whom comes from one of the 28 districts in Malawi, who must be enrolled in a university to guarantee that there are at least 12 who come from the same district? (**3 marks**)

Solution

11×28 (one student is not included to avoid having 12 Student from one district)

$\therefore 11 \times 28 = 308$

The next student has to be included thus $308 + 1 = 309$ Students.

Question 2

(a) A lecturer at Malawi University of Science and Technology wants to assign 5 students to different project topics selected from 8 available topics. How many ways can the topics be assigned if each student must receive a different topic? (**4 marks**)

Solution

To calculate number of ways to assign the topics (Permutation)

Number of ways $= \frac{n!}{(n-r)!}$

$$= \frac{8!}{(8-5)!} = \frac{8!}{3!}$$

$$= \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1}$$

$$= 8 \times 7 \times 6 \times 5 \times 4$$

$$= 6720 \text{ ways}$$

(b) How many license plates can be made using either two letters followed by three digits or two digits followed by four letters? (**4 marks**)

Solution

License plates: (2 letters + 3 digits or 2 digits + 4 letters)

There are 26 letters and 10 digits (0-9).

Case 1 (2 letters + 3 digits)

$$= 26 \times 26 \times 10 \times 10 \times 10$$

$$= 26^2 \times 10^3$$

$$= 676,000$$

Case 2 (2 digits + 4 letters)

$$
\begin{aligned}
&= 10 \times 10 \times 26 \times 26 \times 26 \times 26 \\
&= 10^2 \times 26^4 \\
&= 45,697,600
\end{aligned}
$$

Total number of plates

$$
\begin{aligned}
&= 676,000 + 45,697,600 \\
&= 46,373,600 \text{ plates}
\end{aligned}
$$

(c) A particular brand of shirt comes in 6 colours, has a male version anda female version, and comes in three sizes (small, medium, large) for each sex. How many different types of this shirt are made? (**5 marks**)

Solution

To calculate the total number of different types of shirts, we use product rule. There are 6 colors, 2 version options (M/F) and 3 size options (small, medium, large).

$$
\begin{aligned}
&= 6 \times 2 \times 3 \\
&= 36 \text{ different types of shirt.}
\end{aligned}
$$

Question 3

(a) Find the number of Mathematics students at College taking at least one of the Languages, French ,German and Russians, given the following data: (**4 marks**)

65 study French	20 Study french and Germany
45 study Germany	25 study French and Russian
42 study Russian	15 study Germany and Russian
	8 study all three languages

Solution

Application of the principle of Inclusion-Exclusion of three sets

$$\begin{aligned} |F \cup G \cup R| &= |F| + |G| + |R| - |F \cap G| - |F \cap R| - |G \cap R| + |F \cap G \cap R| \\ &= 65 + 45 + 42 - 20 - 25 - 15 + 8 \\ &= 65 + 45 + 42 + 8 - 60 \\ &= 160 - 60 \\ \therefore |F \cup G \cup R| &= 100 \text{ Mathematics students} \end{aligned}$$

(b) How many bit of strings are there of length eight, either start with bits **000** or end with bits **11**? (**5 marks**)

Solution

8-bit strings starting with 000 OR ending with 11 using the principle of inclusion and exclusion

$$
\begin{aligned}
\text{count strings with 000} &= 8-3=5 \text{ bits}\\
2^5 &= 32 \text{ bits of strings}\\
\text{Count strings with } 11 &= 8-2=6 \text{ bits}\\
2^6 &= 64 \text{ bit of strings}\\
\text{count strings that start with 000 and end with } 11 &= 8-3-2=3\\
2^3 &= 8\\
\text{Total number of bit of strings} &= (32)+(64)-8\\
&= 96-8\\
&= 88 \text{ bit strings.}
\end{aligned}
$$

(c) A network administrator at Malawi University of Science and Technology needs to assign unique login IDs to 3 new students from a pool of 6 available IDs. In how many ways can the login IDs be assigned if no two students can share the same ID? (**5 marks**)

Solution

To assign unique login IDs to 3 new students from a pool of 6 available IDs (permutation):

$$
\begin{aligned}
P(6,3) &= \frac{n!}{(n-r)!} = \frac{6\times5\times4\times3\times2\times1}{3\times2\times1}\\
&= \frac{6!}{(6-3)!}\\
&= \frac{6!}{3!}\\
&= 6\times5\times4\\
&= 120 \text{ ways}
\end{aligned}
$$

(d) A database system at Malawi University of Science and Technology must assign unique usernames to 6 students from a list of 10 available usernames. Each username can be assigned to only one student, and no two students are allowed to share the same username. How many different ways can the system assign the usernames to the students? (**5 marks**)

Solution

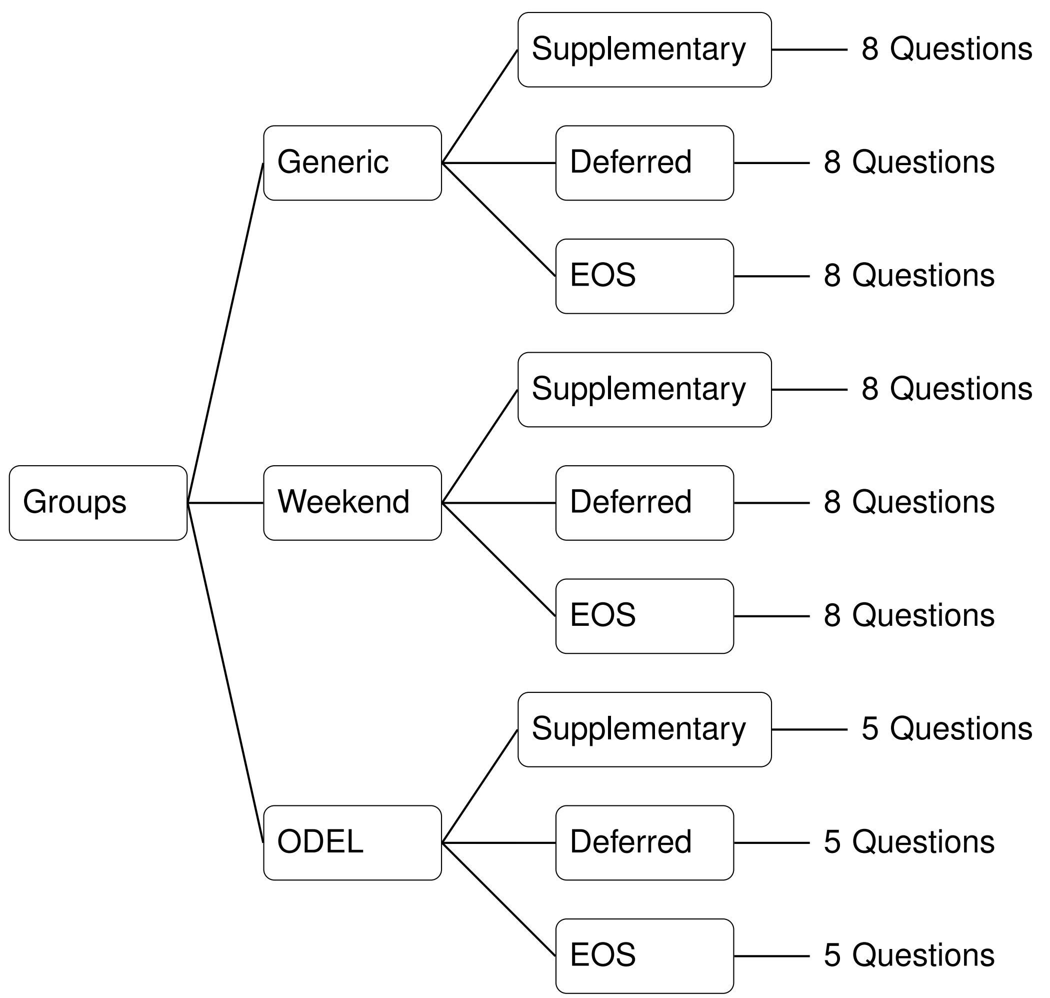
$$
\begin{aligned}
\mathrm{P}(n,r) &= \frac{n!}{(n-r)!} \\
\text{where } n &= 10 \text{ (total usernames)} \\
r &= 6 \text{ (number of students)} \\
&= \frac{10!}{(10-6)!} \\
&= \frac{10!}{4!} \\
&= \frac{10\times 9\times 8\times 7\times 6\times 5\times 4\times 3\times 2\times 1}{4\times 3\times 2\times 1} \\
&= 151,200 \text{ ways}
\end{aligned}
$$

Question 4

(a) The Department of Mathematics at the Malawi University of Science and Technology prepares examinations for three different student groups: ODEL, Weekend, Generic.For each group, the department may set one of three types of examination papers: EOS (End of Semester), Deferred, Supplementary.Each paper consists of a fixed number of questions. The Weekend and Generic groups each receive papers containing 8 questions, while the ODEL group receives papers containing 5 questions. Assume that: Each combination of group and paper type represents a distinct examination.Each question on a paper is considered distinct and contributes to the structure of that examination.

i. Draw a tree diagram to represent the possible examination structures. (**6 marks**)

Solution



ii. Determine the total number of distinct examination structures that can be produced. (**4**

marks)

Solution

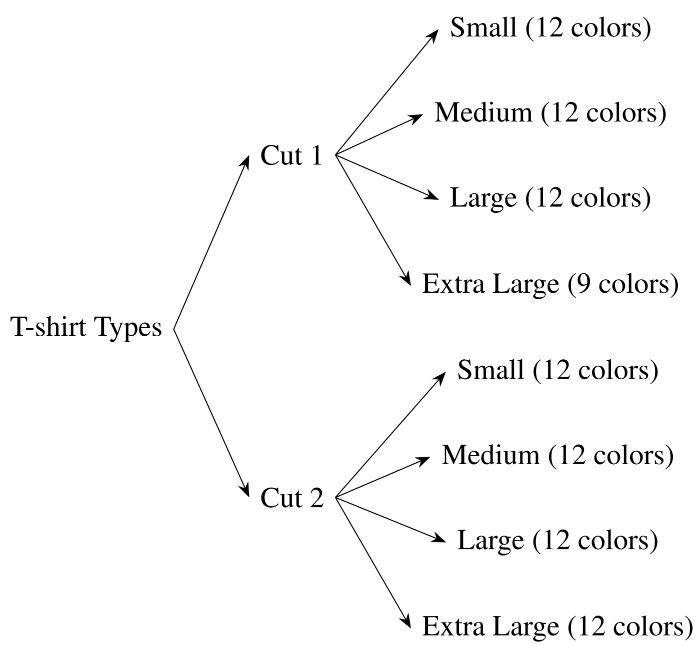
To determine the total number of distinct examination structures that can be produced

$$\begin{aligned}
\text{Structures for ODEL} &= 3 \times 5 && = 15 \\
\text{Structures for Deferred} &= 3 \times 8 && = 24 \\
\text{Structures for Supplementary} &= 3 \times 8 && = 24 \\
\therefore \text{Total} &= 15 + 24 + 24 \\
& && = 63
\end{aligned}$$

(b) A CSS & BIT club at Malawi University of science and technology produces a "Discrete Mathematics" T-shirt for an upcoming academic conference.The shirt is manufactured in two distinct cuts, each tailored to a different body design. For each cut, the shirt is available in four sizes: small, medium, large, and extra large. All sizes are produced in 12 distinct colours, except the extra-large size for one cut, which is only available in 9 colours due to fabric limitations.Assuming that every choice of cut, size, and colour determines a unique product type, determine the total number of different T-shirt types the company produces.

i. Draw a tree diagram to represent all possible types of T-shirts produced. Clearly indicate the number of colour options at the appropriate branches. (**6 marks**)

Solution



ii. Using appropriate counting principles, determine the total number of different T-shirt types produced. Show all working. **(4 marks)**

Solution

The product Rule and sum Rule are used

Cut 1 = Small + Medium + Large + Extra Large

$$= 12 + 12 + 12 + 9$$
$$= 45$$

Cut 2 = Small + Medium + Large + Extra Large

$$= 12 + 12 + 12 + 12$$
$$= 48$$

Total T-shirts types = 45 + 48

Total T-shirt types = 93

Question 5

(a) Find the number of bit strings of length 4 with no consecutive 1s. (**5 marks**)

Solution

$$\text{Bit strings of length 4 with no consecutive 1s}$$
$$\text{Case 1 0 ones: } 0000 \text{ (1 string)}$$
$$\text{Case 2 1 ones: } 1000, 0100, 0010, 0001 \text{ (4 strings)}$$
$$\text{Case 3 2 ones (non-consecutive): } 1010, 1001, 0101 \text{ (3 strings)}$$
$$\text{Total number of strings: } 1 + 4 + 3 = 8$$

(b) The CSS & BIT Club at Malawi University of Science and Technology is organizing a "Discrete Mathematics Challenge" for students.The challenge consists of digital learning modules, created in two formats: Interactive and Text-based.Each module format is available in four difficulty levels: Beginner, Intermediate, Advanced, and Expert. All difficulty levels are designed with 15 distinct themes, except the Expert level, which is available in only 12 themes due to content limitations.Assume that every combination of format, difficulty level, and theme represents a unique module.Using a systematic method to organize the choices at each stage, determine the total number of distinct modules produced. (**9 marks**)

Solution

To calculate total number of distinct modules produced
Formats: 2 (Interactive and Text-based)
Levels: 4 (Beginner, Intermediate, Advanced, Expert)
Themes: Beginner, Intermediate and Advanced has 15 themes each and Expert has 12 themes.
Distinct modules are:

$$\begin{aligned}
\text{Beginner} &= 2 \times 15 = 30 \\
\text{Intermediate} &= 2 \times 15 = 30 \\
\text{Advanced} &= 2 \times 15 = 30 \\
\text{Expert} &= 2 \times 12 = 24 \\
\text{Total} &= 30 + 30 + 30 + 24 \\
&= 114 \text{ distinct modules.}
\end{aligned}$$

(c) A CSS & BIT online portal requires students to create access codes that are between 4 and 6 characters long, where each character can be either an uppercase letter (A–Z) or a digit (0–9). Each access code must contain at least one letter and at least one digit, so codes consisting entirely of letters or entirely of digits are not allowed. Repeating characters are permitted. Determine the total number of valid access codes that satisfy these conditions, taking into account all possible lengths, and show all your working. (**6 marks**)

Solution

Valid access codes (Length 4 to 6)

26 letters + 10 digits possibilities = 36 possibilities

For a code of length n:

- Total possible codes = 36^n
- Invalid (All letters) = 26^n
- Invalid (All digits) = 10^n

Valid codes = $36^n - (26^n + 10^n)$

Length 4:

$$\begin{aligned} 36^4 - (26^4 + 10^4) &= 1,679,616 - (456,976 + 10,000) \\ &= 1,212,640 \end{aligned}$$

Length 5:

$$\begin{aligned} 36^5 - (26^5 + 10^5) &= 60,466,176 - (11,881,376 + 100,000) \\ &= 48,484,800 \end{aligned}$$

Length 6:

$$\begin{aligned} 36^6 - (26^6 + 10^6) &= 2,176,782,336 - (308,915,776 + 1,000,000) \\ &= 1,866,866,560 \end{aligned}$$

Total Valid codes:

$$\begin{aligned} &= 1,212,640 + 48,484,800 + 1,866,866,560 \\ &= 1,916,564,000 \end{aligned}$$

Question 6

(a) At Malawi University of Science and Technology, a new digital learning platform requires each student to create a secure access profile consisting of the following components: A faculty code selected from the set $F = \{$ICT ,Engineering, Science , Business$\}$; A programme code selected from a set P containing 6 programmes; A year of study selected from the set $Y = \{1, 2, 3, 4\}$; A tutorial group selected from a set T containing 5 groups. Each student profile is represented as an ordered quadruple $(f, p, y, t) \in F \times P \times Y \times T where f \in F, p \in P, y \in Y, t \in T$

i. Determine the number of possible student profiles that can be in the system.(**5 marks**)

Solution

$$\begin{aligned}
\text{Faculty codes: } |F| &= 4 \\
\text{Programme codes: } |P| &= 6 \\
\text{Year of Study : } |Y| &= 4 \\
\text{Tutorial groups: } |T| &= 5 \\
|F \times P \times Y \times T| &= 4 \times 6 \times 4 \times 5 = 480 \text{ Profiles}
\end{aligned}$$

ii. Suppose the university introduces an additional set S containing 3 study modes: $\{$Full-time,Weekend,Distance$\}$. Each student profile is now represented by $(f, p, y, t, s) \in FPYTS$. Determine the new number of possible profiles. (**5 marks**)

Solution

With study modes added ($S = 3$)

$$\begin{aligned}
|F| &= 4 \\
|P| &= 6 \\
|Y| &= 4 \\
|T| &= 5 \\
|S| &= 3
\end{aligned}$$

$|F \times P \times Y \times T \times S| = 4 \times 6 \times 4 \times 5 \times 3 = 1,440$ Profiles.

(b) At Malawi University of Science and Technology, the CSS & BIT Club organizes a Digital

Skills Boot camp consisting of several specialized training tracks.Students may enroll in exactly one of the following tracks: A1: Programming Fundamentals (48 students); A2: Data Analytics (36 students); A3: Cybersecurity Basics (41 students); A4: Web Development (35 students). Assume that no student is allowed to enroll in more than one track.

i. Explain why the sets A_1, A_2, A_3, A_4 are pairwise disjoint. (**2 marks**)

Solution

It is because each student enrolls in exactly one track, no student belongs to more than one set and their intersections are empty.

ii. Using an appropriate counting principle, determine the total number of students participating in the boot camp. (**3 marks**)

Solution

Total number of students

$A_1 \quad = 48$ students

$A_2 \quad = 36$ students

$A_3 \quad = 41$ students

$A_4 \quad = 35$ students

$$\begin{aligned}|A_1| + |A_2| + |A_3| + |A_4| &= 48 + 36 + 41 + 35 \\ &= 160 \text{ students.}\end{aligned}$$

iii. Suppose later it is discovered that 5 students accidentally registered in both Programming Fundamentals and Web Development due to a system error.Explain why the counting method used in part (ii) would now give an incorrect result, and state what issue arises when the sets are no longer disjoint. (**3 marks**)

Solution

With 5 Students double-counted (in both A1 and A4) The method in (ii.) is incorrect becouse the sets are no longer disjoint. The issue is double-counting due to overlapping memberships and can be corrected using the principle of inclusion and exclusion

The End of the Question Paper